Task 2: Curve-Inversion Recession Signal

Data and Methodology:

I downloaded monthly US Treasury constant-maturity yields from FRED and built two spreads: the 2s10s spread (GS10 - GS2, available from June 1976) and the 3m10y spread (GS10 -TB3MS, available from April 1953). Official US recession dates come from the NBER recession indicator (USREC), and monthly stock and T-bill returns come from the Ken French data library (US market total return, Mkt-RF + RF, and the risk-free rate RF).

An inversion episode is a run of consecutive months with a negative spread; re-inversions within 12 months are merged into one episode. For each episode I measured the number of months until the next recession started, both from the start of the inversion and from the un-inversion. An inversion followed by a recession within 24 months counts as a true signal; otherwise it is a false signal.

I then compared two rules with buy-and-hold stocks. Rule A moves the portfolio from stocks to T-bills when the curve inverts. Rule B waits until the curve is no longer inverted. In both rules the portfolio stays in T-bills for 18 months and then returns to stocks. The signal is observed at the end of month t and traded in month t+1 to avoid look-ahead bias, and the rules use only yield data, not NBER dates, which are announced with a long delay.

Results:

1) Signal reliability.

Most inversions were followed by a recession: 4 of 6 for the 2s10s spread and 6 of 8 for the 3m10y spread. The 1998 false signal was a one-month inversion of only -2 bp, and the 1966 inversion preceded a recession only after 40 months. The most important false signal is the latest one: in 2022-24 both spreads recorded their deepest inversions in decades (-0.93 pp and -1.57 pp), yet no recession had started by September 2026. The depth of an inversion therefore does not predict whether a recession follows, possibly because QE has compressed the term premium. The signal also missed some recessions: 2s10s did not invert before the 2020 COVID recession, an external shock, and 3m10y stayed slightly positive before the 1957, 1960 and 1990 recessions.

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=== 2s10s (sample from 1976-06-01) ===
Inversion episodes: 6 | followed by recession within 24m: 4 | false signals: 2
Recessions not preceded by an inversion: [2020-03-01]
inv_start   inv_end     months  min_spread  next_rec    lead_inv  lead_uninv  true_signal
 1978-09-01  1982-06-01      46       -2.13  1980-02-01      17.0       -29.0         True
 1989-01-01  1990-03-01      15       -0.32  1990-08-01      19.0         4.0         True
 1998-06-01  1998-06-01       1       -0.02  2001-04-01      34.0        33.0        False
 2000-02-01  2000-12-01      11       -0.41  2001-04-01      14.0         3.0         True
 2006-02-01  2007-05-01      16       -0.14  2008-01-01      23.0         7.0         True
 2022-07-01  2024-08-01      26       -0.93        None       NaN         NaN        False

=== 3m10y (sample from 1953-04-01) ===
Inversion episodes: 8 | followed by recession within 24m: 6 | false signals: 2
Recessions not preceded by an inversion: [1957-09-01, 1960-05-01, 1990-08-01]
inv_start   inv_end     months  min_spread  next_rec    lead_inv  lead_uninv  true_signal
 1966-09-01  1967-01-01       5       -0.34  1970-01-01      40.0        35.0        False
 1969-01-01  1970-01-01      13       -0.29  1970-01-01      12.0        -1.0         True
 1973-06-01  1974-09-01      16       -1.27  1973-12-01       6.0       -10.0         True
 1978-12-01  1981-08-01      33       -2.65  1980-02-01      14.0       -19.0         True
 2000-08-01  2000-12-01       5       -0.53  2001-04-01       8.0         3.0         True
 2006-08-01  2007-04-01       9       -0.38  2008-01-01      17.0         8.0         True
 2019-06-01  2020-02-01       9       -0.32  2020-03-01       9.0         0.0         True
 2022-11-01  2024-11-01      25       -1.57        None       NaN         NaN        False
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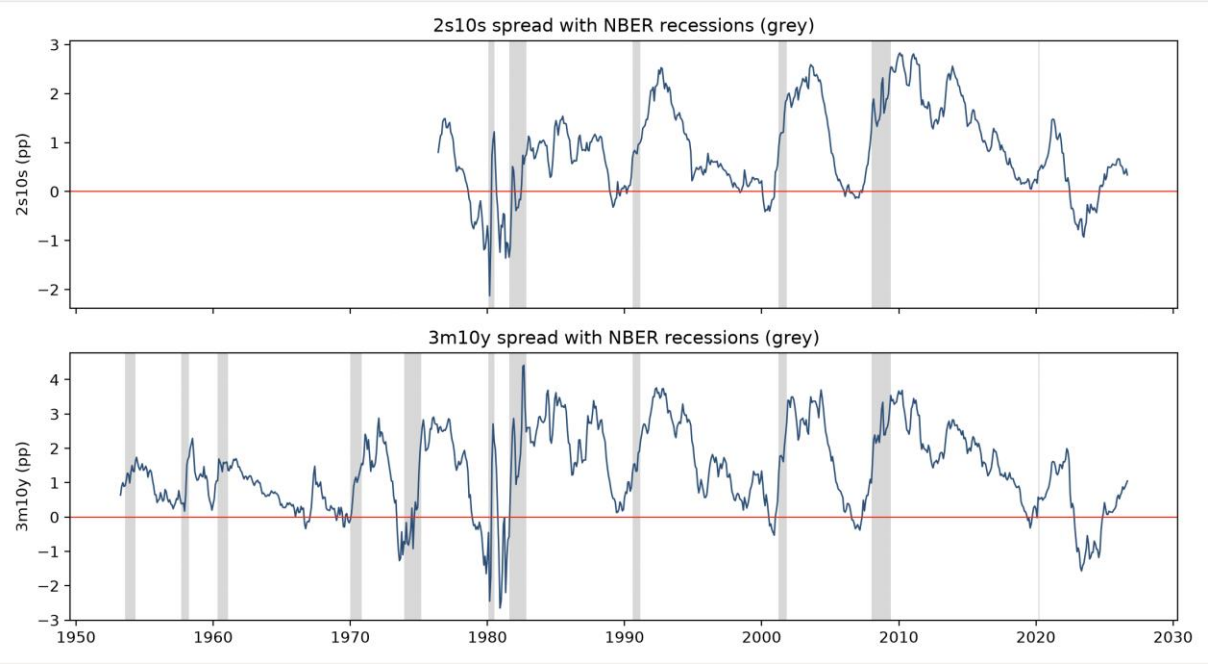


Figure 1. 2s10s and 3m10y spreads with NBER recessions (grey)

2) Lead time.

For true signals, the lead time from the start of the inversion to the recession ranged from 14 to 23 months (mean 18.2) for 2s10s and from 6 to 17 months (mean 11.0) for 3m10y, so the signal says a recession is likely without saying when. Measured from the un-inversion, the lead time clusters around zero (mean −3.8 and −3.2 months): in 1990, 2001, 2008 and 2020 the recession started 0–8 months after the curve steepened again, while negative values (1973, 1980) mean the recession began while the curve was still inverted.

```
2s10s  Lead time from inversion (true signals):    mean 18.2, min 14, max 23 months
       Lead time from un-inversion (true signals): mean -3.8, min -29, max 7 months
3m10y  Lead time from inversion (true signals):    mean 11.0, min 6, max 17 months
       Lead time from un-inversion (true signals): mean -3.2, min -19, max 8 months
```

3) Trading rules.

Both rules reduce volatility from about 15% to 13%, since they hold T-bills 16-18% of the time, but neither dominates. With the 2s10s signal, Rule B performs best (Sharpe 0.63 vs 0.55 for buy-and-hold): it moved into T-bills in mid-2007 and avoided most of the 2008 crash, reducing the maximum drawdown from -50.3% to. -29.9%. With the 3m10y signal the ranking reverses: Rule A earns the highest return (12.0%) and Sharpe ratio (0.63), while Rule B does no better than buy-and-hold because it stayed invested through the 1973-74 bear market. With only 6-8 episodes, a single well- or badly-timed exit drives each comparison.

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=== Rules on 2s10s: 1976-06-01 to 2026-09-01, hold 18m in T-bills ===
                    Ann. return  Ann. vol  Sharpe  Max drawdown  % months in bills
Buy & hold stocks         0.121     0.154   0.553        -0.503              0.000
Rule A (invert)           0.119     0.136   0.590        -0.503              0.179
Rule B (un-invert)        0.124     0.133   0.632        -0.299              0.179

=== Rules on 3m10y: 1953-04-01 to 2026-09-01, hold 18m in T-bills ===
                    Ann. return  Ann. vol  Sharpe  Max drawdown  % months in bills
Buy & hold stocks         0.113     0.150   0.522        -0.503              0.000
Rule A (invert)           0.120     0.131   0.626        -0.459              0.163
Rule B (un-invert)        0.105     0.132   0.516        -0.465              0.163
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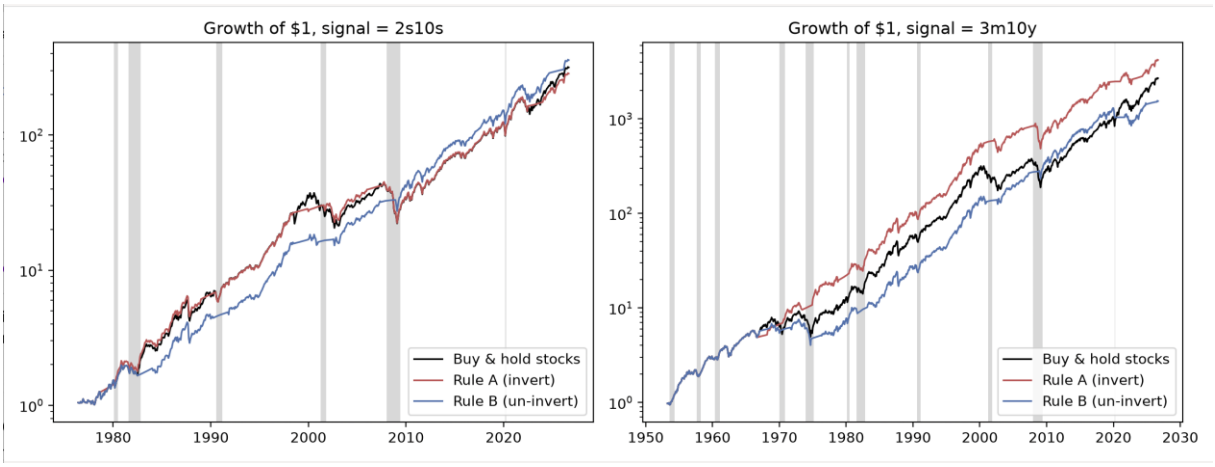


Figure 2. Growth of $1: buy-and-hold vs Rule A and Rule B (log scale)

Conclusion:

The yield-curve inversion has been a useful but imperfect recession signal. It preceded most US recessions, but it produced false signals, most notably in 2022-24, and its lead time varies by more than a year. As a trading rule it modestly improves risk-adjusted returns, but the better rule depends on the spread used and on the timing of a few episodes, so the signal is better used as one input to a macro view than as a mechanical rule.

Limitations: the sample contains few recessions; monthly average yields can miss short inversions (the daily 2s10s briefly inverted in August 2019); TB3MS is quoted on a discount basis while GS10 is bond-equivalent; transaction costs are ignored; and the 18-month holding period and 24-month window are our own choices.